

Container-Limited Support Expression vs. Pre-Saturation Breathing in the HSF Mesoscape Program

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Abstract

Recent exact mesoscape runs have broken the old fill-and-freeze pathology, but they have not yet produced the stronger breathing mode originally hoped for. Instead, the present regime exhibits structured front growth to full occupancy followed by a narrow birth/retire oscillation at capacity. This note clarifies the scientific meaning of that result.

The central ambiguity is the following. Does the current behavior indicate that the constrained wavefunction naturally expands to fill whatever explicit support container is available, with compression becoming relevant only at capacity? Or does it indicate that the present implementation still allows support occupancy to outrun differentiated expression, so that compression turns on too late? These two readings are not equivalent.

This note argues that the current exact results should be interpreted as a *container-limited support-expression regime*, not yet as the final asymptotic substrate regime. It then defines the missing target: *pre-saturation breathing*, meaning a regime in which lawful birth and lawful compression become competitive before total support occupancy is reached. This yields a sharper research question than “does the model churn?”: under what conditions does faithful differentiated expression saturate before support occupancy saturates?

1 Purpose

The mesoscape program has now reached a useful but incomplete stage. The current exact bounded runs can produce:

- structured front growth rather than pathological fan growth,
- lawful retirement rather than permanent freeze,
- a bounded birth/retire cycle rather than terminal occupancy lock.

However, they do *not* yet produce the stronger breathing mode desired for the broader substrate picture. The hoped-for behavior is something like:

the system stabilizes around a mean support level below full capacity, with births and deaths interleaving irregularly and continuously.

Instead, the present exact bounded runs exhibit:

growth to full occupancy, followed by narrow oscillation at or just below capacity.

This note is meant to clarify what that does and does not mean.

2 The current empirical regime

The current exact mesoscale results support the following descriptive summary:

1. Birth is strongly favored during the initial growth phase.
2. Growth proceeds through a structured advancing support front rather than unconstrained fan duplication.
3. Once full occupancy is reached, retirement can become competitive.
4. The resulting late-time behavior is a narrow occupancy oscillation near capacity.
5. Weakening is still not an active winning channel in the observed exact runs.

So the present regime is no longer:

fill and freeze.

But it is not yet:

broad distributed churn below full occupancy.

The correct current description is therefore:

structured front growth plus capacity-edge breathing.

3 The central ambiguity

The difficult scientific question is how to interpret this behavior.

There are two live readings.

3.1 Reading A: container-limited expression of a naturally expansive wavefunction

On this reading, the wavefunction really does tend to express itself into as many explicit supports as the available container permits. The finite exact box functions as a support container, and the wavefunction fills it because:

- more explicit support is initially better local expression,
- the substrate is genuinely generative,
- compression becomes relevant only when redundancy is forced by the finite box.

Then the observed near-capacity oscillation is a finite-container analogue of a more general substrate tendency:

the wavefunction expands into available support until support redundancy becomes unavoidable.

3.2 Reading B: occupancy outruns differentiated expression in the current implementation

On this reading, the present implementation still lets explicit support accumulate too easily. Growth remains favored not because the substrate fundamentally wants full occupancy, but because:

- differentiated expression saturates too late,
- compression pressures become competitive only after occupancy saturation,
- weakening and pre-saturation retirement are still underpowered.

Then the observed near-capacity oscillation is not the correct asymptotic substrate regime, but rather:

a finite-box artifact of a model in which support occupancy saturates before faithful differentiated expression saturates.

4 Why the ambiguity matters

These two readings differ in scientific content.

If Reading A is correct, then the current exact box is already capturing something real and important:

support-filling is a natural intermediate tendency of the constrained wavefunction.

If Reading B is correct, then the present regime is still missing a key feature:

compression should become competitive before total support occupancy is reached.

The distinction matters because the broader HSF picture is not merely permanently generative. It is also permanently compressive. A regime in which compression only becomes meaningful at total occupancy is therefore suggestive but incomplete.

5 Primary interpretive claim

The best current interpretation is:

The present exact mesoscape results should be treated as a container-limited support-expression regime, not yet as the final asymptotic substrate regime.

This is the most cautious and most useful reading because it preserves both insights:

1. the wavefunction may indeed try to express itself broadly into available support,
2. but the current implementation still may not be allowing compression to compete early enough.

So the present results are not best read as:

“the substrate really only does box-filling plus edge oscillation.”

They are better read as:

“in the current exact bounded regime, support expression reaches the container ceiling before a richer breathing mode opens.”

6 What is missing: pre-saturation breathing

The physically desired regime is stronger than the current one.

Define:

Pre-saturation breathing. A mesoscale regime in which lawful births and lawful compression events become mutually competitive before total support occupancy is reached, producing irregular but bounded turnover around a stable mean support level strictly below $n_{\max}$.

This is the regime that would look like:

- some supports die before the box is full,
- some supports are born while others are retired,
- the mean active support count stabilizes below capacity,
- the system breathes without requiring full occupancy first.

That is the missing milestone.

7 Reframing the main research question

The wrong question is:

Does the wavefunction want to fill its container?

The better question is:

Under what conditions does faithful differentiated expression saturate before support occupancy saturates?

That is the right pressure point because it separates:

- support quantity,
- from support quality,
- and occupancy,
- from differentiated local expression.

If differentiated expression saturates first, then births lose relevance and compression can become competitive below capacity. That opens the pre-saturation breathing regime.

If occupancy saturates first, then the system hits the box ceiling before compression matters, and the result is the narrow capacity-edge oscillation currently observed.

8 Conceptual picture

The current picture may therefore be stated as follows.

1. The global wavefunction is primary.
2. Explicit subsystem support is secondary and constraint-driven.
3. The wavefunction may naturally seek broader explicit support when under-expressed.
4. But broad explicit support is not yet the same thing as faithful differentiated expression.
5. The present bounded exact runs seem to reach support saturation before a richer differentiated-breathing regime appears.

Thus the current regime is neither a trivial failure nor a final success. It is a finite-size intermediate behavior.

9 Consequences for interpretation of exact runs

The following interpretive rules are adopted.

1. Fill-to-capacity does not by itself falsify the framework

A bounded exact box may naturally be too small to display the broader substrate regime.

2. Capacity-edge oscillation is meaningful but not sufficient

It shows that the model now has a lawful compression channel, but it does not yet establish broad churn.

3. Weakening remains a missing compression mode

As long as weakening never wins, the model is still relying on discrete support turnover rather than smoother re-expression.

4. The current exact regime should be treated as a finite-size diagnostic regime

Its value lies in showing which pressures are turning on, when, and in what order.

10 What would count as progress beyond the current regime

The following would count as evidence that the model is moving beyond container-limited support expression:

1. first retirement occurs before full occupancy,
2. weakening becomes a winning move in some runs,
3. active support fluctuates around a mean strictly below $n_{\max}$,
4. the turnover set broadens beyond a small boundary family,

5. differentiated expression clearly saturates before occupancy does.

Any of these would strengthen the case for pre-saturation breathing.

11 Failure conditions

This interpretive program would fail if one of the following became clear:

1. all bounded exact runs always show fill-to-capacity followed only by narrow edge oscillation,
2. weakening never becomes relevant even as bookkeeping improves,
3. support turnover remains forever confined to a trivial boundary flicker,
4. the current limit-cycle pattern persists identically across size and seed with no broadening trend.

In that case the current implementation would be better described as a capacity-cycle model than as a model of broad mesoscopic churn.

12 Immediate research consequence

The next goal should not be phrased as:

make the model churn somehow.

It should be phrased as:

find conditions under which differentiated expression saturates before total occupancy does.

That is the correct scientific target because it is specific, interpretable, and directly tied to the physical meaning of the expression term.

13 Conclusion

The current exact mesoscale results do not yet justify the claim that the model exhibits broad substrate breathing below capacity. They also do not justify the claim that the substrate fundamentally prefers only to fill its container and bounce.

What they do justify is a more limited and more useful statement:

The current exact bounded regime is a container-limited support-expression regime: structured front growth proceeds until support capacity is reached, and only then does a narrow birth/retire breathing mode appear.

The missing milestone is now clear:

pre-saturation breathing, in which lawful compression becomes competitive before total support occupancy is reached.

That is the right target for the next stage of the mesoscale program.